

# Liquid-Liquid Extraction

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# Introduction

- Using a 3-stage mixer-settler system to extract acetic acid from White Oil using water as a solvent
- The independent variable will be the three chosen mixer speeds and the dependent variable will be the efficiency of the extraction
- Industrial application
  - Essential oil extraction for pharmaceuticals
  - First step in recovery and purification of bioprocess products
  - Removal of organics like phenol from wastewater

# Theory

Liquid-liquid extraction is a way to separate compounds based on relative solubilities in two immiscible liquids.

A solute is introduced and creates a new phase partially miscible with the liquid to be extracted

The transfer of species from one liquid to the other liquid is driven by chemical potential

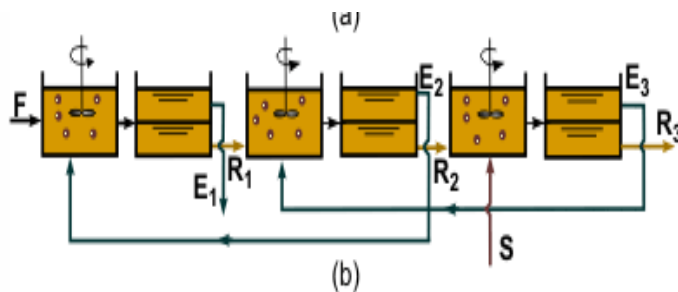
“Once the transfer of species is complete, the overall system of chemical components that make up the solutes and the solvents are in a more stable configuration.”

The product which contains the solvent is called the “extract”. The residual liquid from which the solute was removed is the raffinate

The separation of the liquids is optimally achieved in a cascade of stages with countercurrent flow of solvent and raffinate with solute

After the two liquids come in contact in one stage with a certain residence time, the raffinate from that stage moves on to another stage with a fresh quantity of solvent to continue the liquid-liquid extraction.

The raffinate continues to a new stage until the solute is removed to the extract.



# Calculation

Murphree efficiency in the extract phase.

$$E_{ME,s} = \frac{Y_{s+1} - Y_s}{Y_{s+1}^* - Y_s}$$

Where

$E_{ME,s}$  = Murphree efficiency in the extract phase for stage  $s$  (%)

$Y_{s+1}^*$  = composition of extract phase in equilibrium with raffinate phase

Murphree efficiency in the raffinate phase.

$$E_{MR,s} = \frac{X_s - X_{s+1}}{X_s - X_{s+1}^*}$$

Where

$E_{MR,s}$  = Murphree efficiency in the raffinate phase for stage  $s$  (%)

$X_{s+1}^*$  = composition of raffinate phase in equilibrium with extract phase

- the overall efficiency.

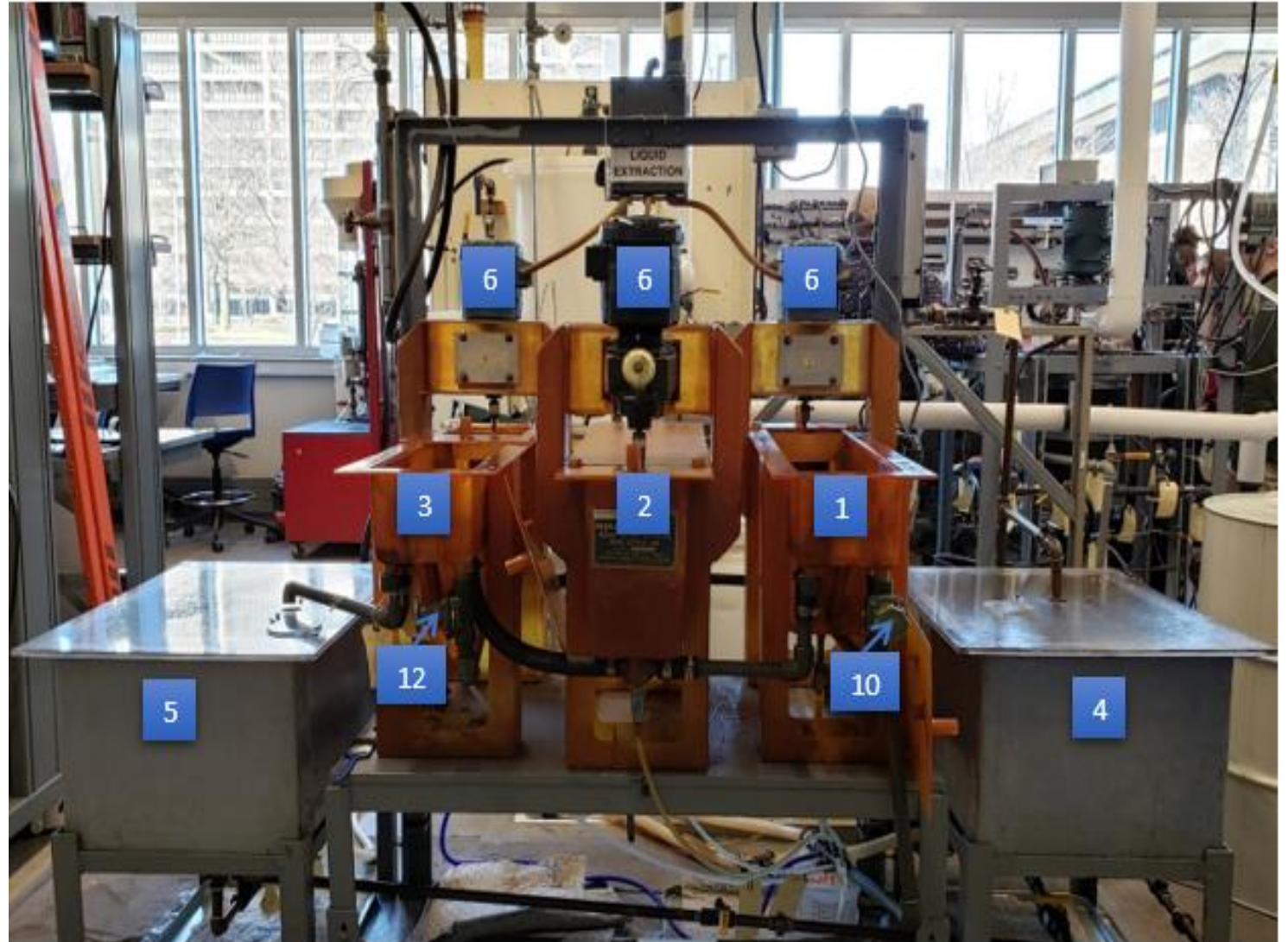
- $E_o = \frac{n_T}{n_{act}}$

- Where

- $E_o$  = overall efficiency (%)
- $n_T$  = number of theoretical stages obtained from McCabe Thiele diagrams
- $n_{act}$  = actual number of stages

# Apparatus

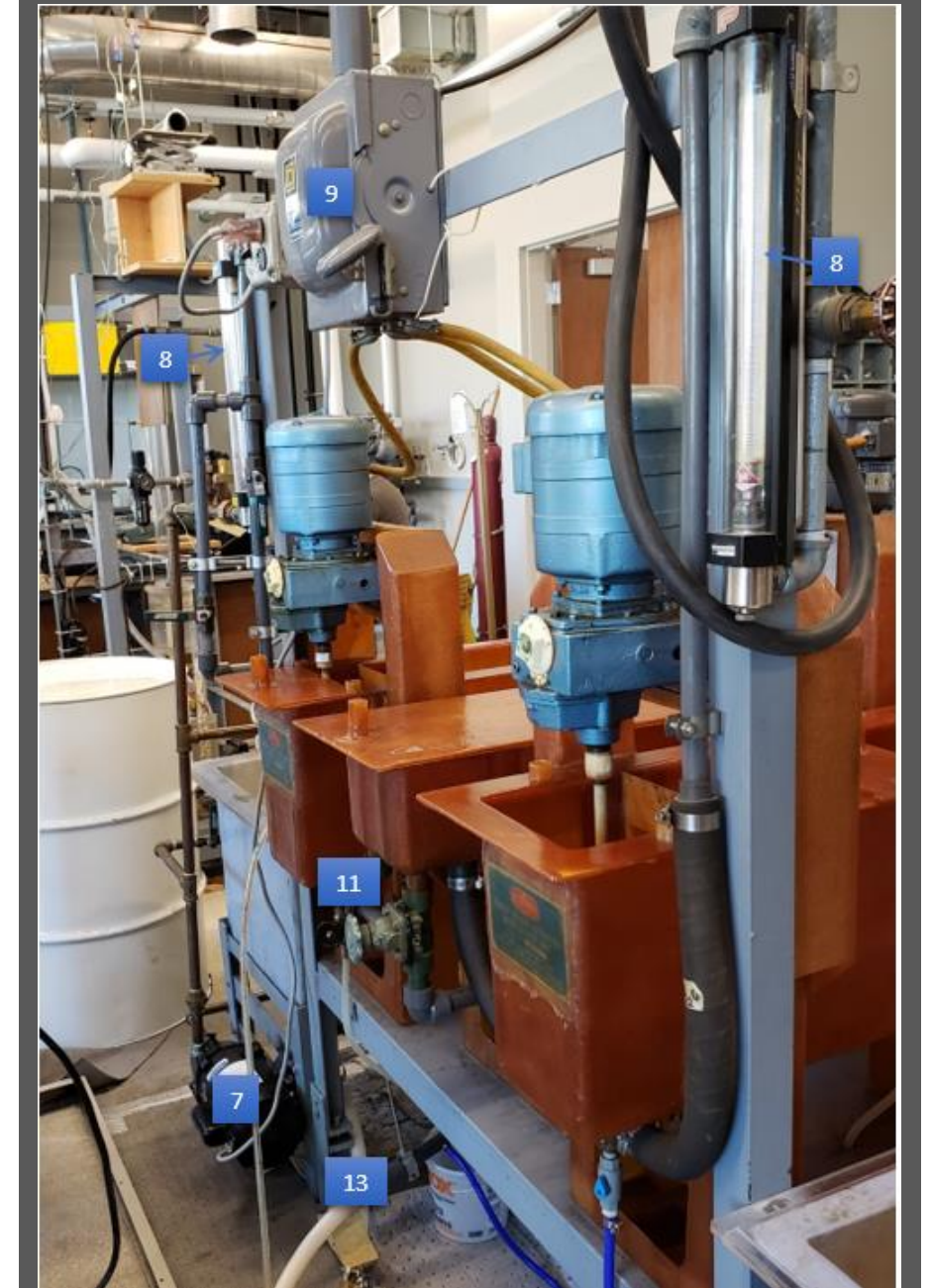
| No. | Component               |
|-----|-------------------------|
| 1   | Stage 1 Extraction Tank |
| 2   | Stage 2 Extraction Tank |
| 3   | Stage 3 Extraction Tank |
| 4   | Extract Tank            |
| 5   | Raffinate Tank          |
| 6   | Motor with Agitator (3) |
| 7   | Motor Pump              |
| 8   | Rotameter (2)           |
| 9   | Safety agitator         |
| 10  | Stage 1 Valve           |
| 11  | Stage 2 Valve           |
| 12  | Stage 3 Valve           |
| 13  | Extract Tank Valve      |





# Material & Supplies

| No. | Component                        |
|-----|----------------------------------|
| 1   | Acetic acid                      |
| 2   | White oil                        |
| 3   | Water                            |
| 4   | NaOH                             |
| 5   | Graduated cylinder               |
| 6   | Beaker                           |
| 7   | Burette and burette standard     |
| 8   | Flask                            |
| 9   | Magnetic Stir bar and stir plate |
| 10  | Volumetric flask                 |
| 11  | Phenolphthalein                  |





# Procedure

## *Pre-lab Day:*

- Study the equipment and get familiar with how it is used.
  - White oil from morning lab period should still be in the feed or product tank. Use this to set up the oil/water extraction circuit.
- Do not drain the oil at the end of the period.

## *Lab Day:*

- Prepare a mixture of 25 gallons of White Oil and acetic acid mixture so that it is 0.5 wt.% acetic acid.
- Mix well 0.361 L of acetic acid and 94.274 L of White Oil in the feed tank by using the centrifugal pump in a closed circuit.
- Take a sample of the well-mixed mixture to determine the composition. ( titrate with standardized 0.05 N NaOH)

- Establish the water circuit in the mixer-settler unit so that steady levels are reached in each stage.
- Adjust the speed of the agitator to 500 RPM.
- Begin feeding White Oil and adjust the flow to a ratio of solvent to feed flow rates of at least 2.0.
- Allow system to run about 5-10 minutes until steady state is established.
- Withdraw a sample of 50 mL of the mixture at each of the three stages.
- Determine the concentration of each sample via titration with NaOH per step 2 of Lab Day procedure.
- Repeat Lab Day procedure from step 4 for agitator speeds of 600 RPM and 700 RPM.
- Keep lid on tank to avoid acetic acid evaporation during experiment.

## Procedure (part 2)





# Experimental challenges

- Titrations were done quickly with water, but not the case with the oil
  - Keep the titrations consistent
- Finding equilibrium data for the McCabe-Thiele diagram.
- Understanding the difference between raffinate and extract
  - Extract is the function of the raffinate.
- Make sure agitator speeds are above 400 RPM and below 1000 RPM

# Safety controls

- Water pump is electrical.
- Glacial acetic acid is corrosive
- NaOH solutions is caustic
- Phenolphthalein is flammable
- shut off switch can be used to turn off pump and experiment
- shut off valve for water
- shut off valve for electrical box